6th Chief Secretaries Conference, 2026

Concept Note for Regular Session on

Technology in Governance

A. Introduction

From the invention of the wheel and the discovery of fire to the steam engine and the printing press, technology has continuously reshaped how societies function and how governments serve their citizens. This continuous evolution of technology has now brought us to a point where it impacts almost every function of government and the lives of citizens, ranging from health service delivery and disease surveillance, digital monitoring for public safety, early warning and response in disaster management, precision advisories in agriculture, to efficient delivery of citizen and welfare services. It spans many technological areas, such as geospatial technology, aerospace and drones, environmental science & technology, biologics & biotechnology, manufacturing and industrial technology, materials science, and marine/ocean technology, each with its own engineering discipline, know-how, history, as well as associated infrastructure and application areas. Illustrative examples across some of these wider technological fields, that have a direct or indirect impact on governance, are set out in the ***Annexure***. States sit at the centre of this transformation, and during the conference, may focus on all such technological interventions which they have introduced to improve governance.

This note focuses on three such impactful technologies:AI and Digital Services, Geospatial technologies, and Microbiome technologies, chosen for their particular potential for their application in better decision making and effective citizen-centric service delivery across States. Government's ability to govern well can significantly improve if the full potential of these three under-governed assets, data, geospatial information, and biological data, each held largely by States, is converted from raw material into working public infrastructure.

1. Data Governance, Government Data for AI, and Practical AI Adoption

- India's first digital decade turned identity, payments and benefit transfer into public infrastructure: Aadhaar, UPI, DigiLocker and DBT have together moved over ₹53 lakh crore to citizens while preventing an estimated ₹5 lakh crore in leakages. That achievement was largely about information technology; the decade ahead turns on a different question, whether government can govern and share its data, whether that data can train Indian AI models, and whether AI gets embedded in the routine work of departments that touch citizens, businesses and employees, and digital services can be localized in the contextual environment.
- Because citizen-facing administration and the transaction-level data beneath every national digital stack sit with States, districts and nearly 5,000 urban local bodies, adoption of this agenda is inevitably uneven across and within States.
- The Centre has already built the enabling layer, the DPDP Act, 2023 and Rules (November 2025), the National Data Governance Committee, the IndiaAI Mission

with AIKosh and sovereign models, and the India AI Governance Guidelines. The focus should now be on converting that scaffolding into working data governance, real contributions to Indian AI, and measurable AI applications on the ground.

2. Geospatial Technology and Governance

- Geospatial information plays an equally foundational role in governance, underpinning planning and decision-making across development, infrastructure, land, environment and disaster management, and States are its principal creators and users.
- Geo-mapping significantly amplifies the value of other datasets. Crime and cyber-crime records, when mapped geospatially, reveal patterns and hotspots that are invisible in tabular data alone, enabling targeted policing and resource deployment. The real gain lies in interconnectedness: overlaying crime, infrastructure, land-use, population and environmental datasets on a common geospatial base allows States to identify correlations and plan interventions that no single dataset, viewed on its own, could reveal.
- The Government of India has already built the policy framework for this, the Geospatial Guidelines, 2021 and the National Geospatial Policy, 2022, aimed at wider use and sharing of geospatial information and a stronger domestic geospatial ecosystem.
- This framework needs to be converted into a coordinated State-level approach: clear institutional responsibility, better information sharing, common approaches, and the capacity to put geospatial information to full use in planning and decision-making. This session is meant to chart a practical roadmap for States to harness that potential for better governance, improved services, and new solutions to development challenges.

3. Microbiome Technologies for Governance

- A third, far less visible asset is biological. Microbiomes, the communities of microorganisms and their genes, products and interactions, form an invisible foundation of human, animal, plant and environmental health, shaping nutrition, immunity, disease, soil fertility, crop and livestock productivity, water quality, pollution degradation, carbon cycling and antimicrobial resistance.
- Climate change, pollution, chemicals, wastewater and land-use change are already disrupting these microbial systems and degrading air, water and soil quality, with consequences that reach into health, food security, livelihoods, economic productivity and ecological stability.
- Microbiome engineering holds transformative potential in health, agriculture and environmental sustainability. India should therefore establish a Microbiome Governance Framework within its One Health and climate-resilience agenda, in support of the Sustainable Development Goals, particularly hea

- lth, nutrition, clean water, climate action, sustainable agriculture and biodiversity, and in furtherance of Viksit Bharat through stronger public health, resilient food and water systems, environmental sustainability, and bio-economy-led development. Such a framework would make critical biological changes measurable and actionable, while protecting biodiversity, privacy, community rights and public trust.

B. <u>Current Situation</u>

Across all three streams, the pattern repeats: institutional frameworks and enabling technology already exist, but the underlying data remains fragmented, and adoption on the ground is real but patchy.

1. Data Governance, Government Data for AI, and Practical AI Adoption

- ***Data governance.*** A National Data Governance Committee, co-chaired by Secretary (Coordination) Cabinet Secretariat, Secretary MeitY and Secretary MoSPI, now guides data sharing within and outside government through Data Sharing Committees in every Ministry/Department, a Model Data Sharing Framework covering dataset inventory, classification and monetisation, and exchange platforms, API Setu, DigiLocker and Entity Locker, complete with approval logs. The DPDP Rules, notified 14 November 2025, give an 18-month compliance window, making every State department holding personal data a data fiduciary. The India Data Management Office under IndiaAI additionally requires Data Management Units in each Department and runs the India Datasets Programme. Even so, on the ground, data remains large but fragmented: sectoral stacks such as AgriStack, ABDM (70 crore+ ABHA accounts), UDISE+/APAAR and e-Courts are building registries, but most State datasets are scheme-specific, vendor-hosted, inconsistently coded and without metadata.
- ***Government data for Indian AI.*** The IndiaAI Mission (₹10,372 crore) has empanelled over 38,000 GPUs at subsidised rates; AIKosh hosts 9,500+ datasets and around 270 models; four sovereign foundation models, Sarvam, BharatGen, Gnani and Soket, were released at the AI Impact Summit in February 2026; Bhashini already covers all 22 scheduled languages; and 570 AI Data Labs and 27 IndiaAI Labs are being set up across States. What is missing is not compute but content: the corpora that would make these models genuinely useful for administration, land and court records, scheme guidelines, grievance texts, field observations, speech in State languages and dialects, sit with States and are not yet available in a lawful, reusable form.The India AI Governance Guidelines, November 2025, set out seven sutras, an AI Governance Group, an AI Safety Institute and a federated incident-reporting mechanism.
- ***Practical AI.*** AI already runs in routine Central use: IGMS within CPGRAMS triages grievances and flags root causes, Income Tax and GSTN run risk analytics, e-Courts uses SUVAS and SUPACE, Kisan e-Mitra answers PM-KISAN queries in eleven languages, IndiaAI is funding 30 sectoral solutions, and 176 AI courses on iGOTKarmayogi have drawn 73 lakh+ enrolments. States, too, have moved from pilots to policy and partnership: Tamil Nadu (a data policy, the Vivasaya Nanban advisory reaching 79 lakh farm households, an AI helpline), Odisha (a data policy, a sovereign

AI partnership, 50 MW of compute), Telangana (open data, analytics and AI frameworks), Maharashtra (MahaAgri-AI), Uttar Pradesh (AI Pragya), Kerala (AI in schools), and Andhra Pradesh (AI and quantum compute hubs). The common pattern across these States is an AI policy, a compute commitment, one flagship citizen-facing use case and a capacity programme for officials.

2. Geospatial Technology and Governance

- ***Institutional adoption by States.*** States have already built institutional mechanisms, GIS cells, State Remote Sensing Agencies, Satellite Application Centres and geoportals, and are increasingly using geospatial information across planning, infrastructure, land, water, agriculture and disaster management.
- ***Geospatial at the core of national programmes***. States are key partners in initiatives such as SVAMITVA, NAKSHA, AMRUT, Namami Gange and the National Hydrology Project, all of which lean on geospatial information to improve planning, monitoring and service delivery.
- ***A common national foundation.*** Survey of India anchors all of this through its geodetic and positioning infrastructure and authoritative topographical data, enabling consistent mapping and integration of geospatial information across States.
- ***Expanding State initiatives.*** Beyond these national programmes, States are running their own diverse geospatial initiatives, supported by national capacity-building and knowledge-sharing efforts, together laying a strong foundation for the next phase, mainstreaming geospatial information into governance across States rather than leaving it in specialised pockets.

3. Microbiome Technologies for Governance

- ***Governance blind spot.*** Public systems already track pollutants, pathogens, soil nutrients, crop yields and disease, but almost none of them monitor the microbial communities that actually link these outcomes together, leaving a gap between what is routinely measured and the underlying biological processes that may contribute to these outcomes.
- ***Fragmented activity.*** India has real but fragmented activity in pathogen and antimicrobial-resistance surveillance, human microbiome research, sustainable agriculture, food regulation, wastewater management and environmental monitoring, spanning health, One Health, agriculture, food, water and environment, but using inconsistent definitions, methods, standards and data systems throughout.
- ***An initial human reference, but only a research resource.*** The DBT-supported Indian Human Microbiome Initiative has generated valuable population-level reference data, but this remains largely confined to research rather than functioning as a routine public-health or nutrition tool.
- ***National One Health Mission*** has been initiated to unite human, animal and environmental health for stronger pandemic preparedness and zoonotic disease control. Steered by DHR–ICMR under the Office of the Principal Scientific Adviser, it brings

together over 16 ministries and departments to shift from reactive response to anticipatory prevention.

- ***The technology is available.*** Metagenomics, biosensors, functional assays, artificial intelligence and biomanufacturing now exist to translate microbial knowledge into public solutions, and Earth observation and environmental data can support the same measurement and translation task.
- ***States are the natural implementation site.*** Because State institutions directly control key delivery systems, health, agriculture, livestock, fisheries, food safety, water, waste and pollution, they are the natural sites for microbiome-enabled governance and demonstration, in exactly the same way they are for geospatial and AI-enabled governance.

C. Challenges

*1. Data Governance, Governme*nt Data for AI, and Practical AI Adoption

- ***Siloed, unlabelled data.*** Scheme-specific MIS run on proprietary schemas, with no master registries for beneficiaries, land parcels, establishments or facilities, and scanned PDFs passing for digitisation, leave no institutional route for sharing data within or outside government.
- ***DPDP uncertainty.*** Most State departments still lack data-protection officers, consent architecture, breach protocols or personal-data inventories, and unclear anonymisation standards mean the default response to any request is to share nothing at all.
- ***No State-level Data Sharing Committee or audit log.*** State-level equivalents of the Data Sharing Committee, the exchange approval mechanism and the audit log largely do not exist, and district and municipal capacity to steward data is thin.
- ***No real channel into Indian AI.*** There is no standard licence, anonymisation benchmark or pipeline for States to contribute corpora to AIKosh, and dialects and tribal languages remain the least represented of all. Much digitised material is unlabelled scans with undocumented provenance and consent status, and States see no clear return for contributing and remain cautious about DPDP liability.
- ***Pilots that do not scale, and thin capacity.*** Partner-run pilots typically sit outside regular workflow and budget; procurement rules are unsuited to outcome-based, pay-per-use AI services; district and block staff are the least trained of all cadres; there is no risk assessment, contestability or audit trail in place before AI touches entitlements, policing or scrutiny; and citizens without smartphones or Hindi/ English literacy risk being excluded altogether.

2. Geospatial Technology and Governance

- ***Limited integration of information.*** Inconsistent coordinate systems, scales, standards and formats across Departments limit the integration, sharing and effective use of geospatial information for comprehensive planning and decision-making.

- ***Limited accessibility and reuse.*** State-generated geospatial data is not always readily available for wider government use, largely due to disputes over ownership and departmental policy restrictions, resulting in duplicated effort and missed opportunities.
- ***Gaps in institutional ownership.*** The absence of clear responsibility for maintaining key datasets affects their currency, reliability, and fitness as a basis for government decisions.
- ***Capacity constraints.*** Geospatial capability remains concentrated in specialised units and has yet to be fully integrated into routine decision-making at the district and local level.
- ***Gap between technology and outcomes.*** New technologies, AI, advanced analytics, sensors, IoT devices, drones and advanced mapping, are increasingly available, but need to be focused on clearly identified State priorities and scaled into sustainable solutions rather than remaining one-off applications.

3. Microbiome Technologies for Governance

- ***Limited biological baselines.*** India currently cannot reliably distinguish normal microbial variation from climate- or pollution-related biological disruption.
- ***Climate-sensitive change going unmonitored.*** Heat, drought, floods, pollution, wastewater and changing land use can all disturb microbial systems, yet such changes are not routinely tracked.
- ***Fragmented institutions and data.*** Human, animal, plant, soil, aquatic and built-environment microbiomes are studied entirely separately, using methods that are not comparable with each other.
- ***Incomplete regulation.*** Current systems do not consistently assess the cumulative effects on microbial communities, ecological persistence, resistance genes, virulence or functional recovery.
- ***Weak translation from research to use.*** Microbial associations are often treated as action-ready evidence, even though the validation, multi-site testing, manufacturing and procurement pathways needed to actually act on them remain limited.
- ***Distinct governance concerns.*** Human microbiome data, microbial strains and associated knowledge require their own safeguards, for consent, privacy, access, benefit sharing, indigenous knowledge, intellectual property and biosafety, that do not arise in the same form for administrative or geospatial data.Given the potential dual-use nature of advanced microbiome research and technologies, appropriate biosecurity, responsible research and dual-use risk assessment should also be incorporated to prevent misuse and protect against emerging biological threats.

D. <u>Possible Solutions</u>

The structural fix is the same in every stream: one accountable State body, common standards adopted once rather than repeatedly, shared national infrastructure built for reuse, and a defined, evaluated menu of applications rather than open-ended pilots.

1. Data Governance, Government Data for AI, and Practical AI Adoption

- ***State data governance.*** Each State should set up a State Data Governance Committee under the Chief Secretary, with a Data Management Unit and data officer in every department, carrying out a State data inventory and classification exercise in line with the Model Data Sharing Framework, adopting MDDS/IndEA/API standards and sectoral IDs such as AgriStack, ABDM and ULPIN, and enabling end-to-end online sharing through API Setu, DigiLocker and Entity Locker; and migration of vendor-locked data to State-owned repositories, with exit clauses in all new contracts.
- ***DPDP readiness.*** States need a time-bound compliance drive, inventories, consent notices, DPOs, breach SOPs and privacy-by-design, built around a model State toolkit and a standard anonymisation protocol issued by MeitY, so that 36 administrations are not left to interpret the Rules 36 different ways.
- ***An 'AIKosh for States' compact.*** States should get a standard licence and pipeline to place anonymised administrative corpora, especially speech and text in State languages, on AIKosh; in return, they should receive priority IndiaAI compute, sovereign models fine-tuned for State languages and departments, and the outputs of the AI Data Labs operating in their State.
- Lighthouse corpora per State, such as land and registration records, court orders, scheme guidelines and GOs, grievance texts and helpline speech, a revenue-sharing principle for commercial use, and a MeitYanonymisation protocol with a DPDP legitimate-use template, so that contributions are lawful by design.
- ***A proven menu of AI use cases.*** Bhashini-based voice assistants for scheme discovery and grievances, IGMS-style grievance triage, AgriStack and satellite-based crop advisory, AI-assisted screening in district hospitals, eligibility and document checks with human sign-off, utility predictive maintenance, and file, order and legislation summarisation for officials, each backed by a reference architecture, procurement template and evaluation protocol.
- ***Enablers.*** A model AI-services tender from MeitY/GeM, pay-per-use, outcome milestones, data and model ownership with the State, exit and audit clauses, the seven sutras of the AI Governance Guidelines as a pre-deployment checklist cleared by a Chief Secretary-chaired committee, and the iGOT AI curriculum for all officers, alongside district 'data stewards' drawn from existing cadres, and voice, vernacular and assisted-access channels designed in from the start.

2. Geospatial Technology and Governance

- ***State-level geospatial governance.*** States should put in place a clear geospatial policy or roadmap, aligned with the Geospatial Guidelines and NGP 2022, and designate a nodal institution to coordinate geospatial activities across departments.
- ***Common standards and a geodetic framework.*** Adopting common standards and a national geodetic reference framework would enable seamless integration and consistent use of geospatial information across departments and States.
- ***Data sharing and custodianship.*** States need clear, accountable arrangements for sharing and maintaining priority datasets, with defined institutional ownership rather than ambiguous responsibility.
- ***Capacity building.*** Geospatial training should be institutionalised across administrative levels and built into government induction, higher education and professional programmes.
- ***Technology matched to State priorities.*** States should identify priority use cases for GeoAI, advanced analytics, Earth observation, Digital Twins and other emerging technologies, and scale only those applications that address specific governance challenges and deliver measurable outcomes.
- ***A Centre-State-Industry partnership.*** States should share priority governance challenges with industry and academia to develop need-based solutions and strengthen the domestic geospatial ecosystem in collaboration with Central Institutions.

3. Microbiome Technologies for Governance

- ***Adopt national One Health framework.*** India should recognise that the planetary microbiome operates as a continuous, interconnected ecological meta-community driving global biogeochemical cycles and regulating ecosystem resilience, and align microbiome action across human health, animals, agriculture, soil, water, air, waste and built environments accordingly.
- ***Sentinel observatories.*** Monitoring microbiome changes across representative populations, agro-climatic zones, cities, river basins and production systems, linked to climate, pollution and health data, would build the biological baseline India currently lacks.
- ***Defined regulatory indicators.*** Microbial indicators for soil and water quality, ecosystem recovery, antimicrobial resistance, air risks and climate-sensitive diseases need to be defined and validated for government use.
- ***Integrated risk and Biosecurity assessment.*** The cumulative and ecological effects of antibiotics, pesticides, chemicals, effluents, land-use change and restoration activities should be assessed proportionately, with appropriate safeguards for responsible research, for eventual embedding into relevant policies and regulations.
- ***A translation pipeline.*** Validated diagnostics, probiotics, bio-fertilisers, biopesticides, climate-resilient agriculture, wastewater treatment, bioremediation and bio-based manufacturing all need a defined route from research finding to deployed product.

- ***Shared national infrastructure.*** An Indian Microbiome Data and Resource Grid should link the Indian Biological Data Centre, reference laboratories, culture collections, AI hubs, bio-foundries and biomanufacturing facilities under common data and quality standards.
- ***State-led pilots.***States should demonstrate measurable benefits in nutrition, enteric disease, antimicrobial resistance, soil restoration, reduced chemical use, livestock health, wastewater treatment and climate-resilient agriculture.

E. <u>Way Forward</u>

Across all three streams, the same phasing logic applies: Central templates and State nodal arrangements in the short term, standards, shared infrastructure and institutional foundation in the medium term, and integrated, budgeted practice in the long term, with reciprocal Centre commitments at each stage.

1. Data Governance, Government Data for AI, and Practical AI Adoption

- **Short-term:**MeitY issues the Model Data Sharing Framework, DPDP toolkit, anonymisation protocol and standard State data licence; every State constitutes its State Data Governance Committee and AI Governance Committee and names nodal officers; and each State notifies a State Data Sharing Policy and State AI Policy, prepares a metadata catalogue of its top 50 datasets, nominates two lighthouse corpora for AIKosh, and selects its 10 flagship AI use cases with a named owner, budget line and outcome metrics.
- **Medium-term:** metadata, data and API standards are adopted; vendor-locked data is migrated to State-owned repositories; 20+ priority datasets are exchanged online through API Setu, DigiLocker and Entity Locker; and two or three lighthouse corpora are placed on AIKosh, with reciprocal compute and fine-tuned models from MeitY; AI Data Labs label dialect and tribal-language material, a fine-tuned State-language model is deployed in at least one department, and a revenue-sharing principle for commercial use is finalised.
- **Long-term:** DPDP compliance is achieved across departments handling personal data, and 10 flagship AI use cases per State, spanning agriculture, grievance redressal, health, education and rural/urban services, are integrated into workflow and budget, with outcomes tracked on a common dashboard; the iGOT AI curriculum is completed by all Group A and B officers, a responsible-AI checklist is applied to every live deployment, and results are presented at the 7thConference along with the annual State ranking on AI adoption.

2. Geospatial Technology and Governance

- **Short-term:** DST issues model SOPs, a service catalogue and a costing template for State drone and remote-sensing capacity; States designate a nodal institution for geospatial activities and prepare an inventory of drones, sensors and geospatial datasets.

- **Medium-term:** States establish a State-level mechanism for sharing and facilitating access to State-generated geospatial information, in line with the Geospatial Guidelines, 2021; designate a nodal department or institution; prepare an inventory of key geospatial datasets and their custodians; identify priority areas for capacity building; and identify 2-3 high-impact geospatial use cases to demonstrate value. States also develop and adopt a State-level geospatial data-sharing framework covering access, custodianship, sharing protocols, standards and safeguards; align priority datasets with the national geodetic reference framework and appropriate metadata; institutionalise capacity building for State and district-level officials; and initiate and scale priority applications, including GeoAI and drones.
- **Long-term:** the use of geospatial information, technology and Decision Support Systems is institutionalised across departments and administrative levels, moving beyond specialised GIS units towards geospatially enabled governance more broadly.

3. Microbiome Technologies for Governance

- **Short-term:** Identify one priority microbial technology opportunity per domain for adoption in governance. DBT and concerned Ministries/Departments should develop model implementation SOPs, assessment criteria and costing templates for priority applications, while States undertake an initial inventory of relevant microbial technology capabilities, infrastructure and implementation gaps.
- **Medium-term:** An interdepartmental expert group formed to map existing programmes, datasets, regulations and infrastructure; identify barriers to bridging the laboratory-to-field gap; and define priority outcomes; initial observatories and State demonstrations selected for purposeful engineering of microbiomes. Common protocols developed; reference laboratories and collections designated; the Indian Microbiome Data and Resource Grid established; and coordinated regulatory guidance developed.
- **Long-term:** longitudinal observatories and multi-State demonstrations are launched across health, agriculture, animals, air, water, soil and climate-sensitive ecosystems, each with a baseline, an accountable department, independent evaluation, and defined continuation criteria. Demonstration of real-world application of microbiome engineering in field.

F. <u>Issues for Deliberation</u>

1. Data Governance, Government Data for AI, and Practical AI Adoption

- The institutional form for State data governance (Committee, Sharing Policy, exchange platforms) should be settled, along with its relationship to the National Data Governance Committee, IDMO, API Setu, DigiLocker, Entity Locker, AIKosh and the Data Protection Board.
- Priority datasets that should be made available through API Setu, DigiLocker and Entity Locker in the first year should be decided, along with the changes in procurement and

contracting norms needed so that data ownership, portability and exit vest with the State in every vendor-built system.

- State datasets should be prioritised for AIKosh and sovereign model training, under a defined licence and anonymisation standard, and what States should receive in return should be settled.
- Whether contribution of anonymised administrative corpora should be entrusted to every State Data Governance Committee should be decided, along with who, AI Data Labs, State universities or Bhashini, should own collection and labelling of dialect and tribal-language speech.
- DPDP obligations should be met in State departmentswithout slowing welfare delivery, with a common template adopted for legitimate public-interest uses where needed.
- The 10 AI use cases every State should commit to should be finalised, with common outcome metrics and a dashboard.
- Safeguards, human oversight, explanation, contestability, incident reporting, should be put in place before AI use in entitlements and enforcement, with a body designated to certify compliance.
- Voice, vernacular and dialect coverage should be built in so that the last mile is reached first, and district and block capacity should be built within existing cadres.

2. Geospatial Technology and Governance

- Responsible sharing and wider use of publicly generated geospatial information should be facilitated, while addressing concerns relating to privacy and ownership.
- An institutional mechanism should be adopted for coordinated management of geospatial activities, including clear responsibilities for priority and authoritative datasets.
- Common standards and a reference framework should be established to ensure consistency, integration, interoperability and effective use of geospatial information across departments and States.
- Geospatial capabilities and technologies should be mainstreamed across departments and administrative levels, beyond specialised GIS units, to support routine planning and decision-making.
- Centre-State-Industry-Academia collaboration should be strengthened to develop, scale and deploy Indian geospatial solutions that address State-specific governance needs.

3. Microbiome Technologies for Governance

- Representative populations, agro-climatic zones, cities, river basins, coasts and climate-vulnerable ecosystems should be selected as priority sites for initial microbiome observatories.
- Indicators of soil, water, ecosystem, antimicrobial-resistance and climate-sensitive health risk that are sufficiently validated for government use today should be identified.

- Common standards for sampling, testing, data management and quality assurance should be set to harmonise implementation across health, agriculture, livestock, water, air, soil and environment programmes.
- The evidence required before microbiome data or products that can inform diagnosis, nutrition, treatment, farming, pollution control or environmental release should be defined.
- The assessment of cumulative ecological effects should be made proportionate, so as to enable safe, low-risk biological technologies and products while still managing genuine risk.
- Safeguards for consent, privacy, biological resources, benefit sharing, intellectual property and indigenous knowledge should be put in place for microbiome-related data and research.
- State-led demonstrations should be selected to show, within 18 months, measurable benefits in nutrition, disease prevention, antimicrobial resistance, soil restoration, water quality, waste management or climate-resilient agriculture.

Annexure–Major Technological Initiatives of the Union Government

Ministry / Department	Technologies	Major Initiatives
Department of Science & Technology (DST)	Quantum computing, communication and sensing; geospatial and geodetic infrastructure; cyber-physical systems	**National Quantum Mission (2023–31, ₹6,003.65 crore):** four thematic hubs (IISc, IIT Madras, IIT Bombay, IIT Delhi); targets of 50–1,000-qubit computers, 2,000 km inter-city quantum key distribution and satellite-enabled quantum links. **Geospatial:** National Geospatial Policy 2022 and Guidelines 2021; Survey of India geodetic/CORS network and authoritative topographic data as the national reference for States. **NM-ICPS and Anusandhan NRF:** 25 Technology Innovation Hubs in robotics, sensors, autonomous systems; NRF as the apex funder for strategic and translational research.

Ministry / Department	Technologies	Major Initiatives
Ministry of Electronics and Information Technology (MeitY)	Digital public infrastructure, unified citizen services, AI access localization	**Aadhaar:** a 12-digit biometric-based unique identity issued by UIDAI (a statutory authority under MeitY), with over 135 crore numbers issued, underpinning identity verification and benefit delivery across government schemes. **UMANG:** a unified mobile application built with the National e-Governance Division, giving citizens a single interface to 2,000+ services from over 100 central and State departments. **Bhashini:** the National Language Translation Mission, launched July 2022, providing AI-driven translation and speech services across the 22 Scheduled Languages, with over 1,000 pre-trained language models available through open APIs. **DigiLocker:** a digital document wallet built by the National e-Governance Division, with over 100 million users and 2,000+ document issuers; documents issued through it carry legal validity equivalent to physical originals under the IT Act. **India Semiconductor Mission:** a ₹76,000 crore programme approved in 2021 to build domestic semiconductor and display fabrication capacity, with multiple fabs and assembly/testing units approved since, aimed at reducing India's reliance on imported chips. **Common Services Centres (CSC) scheme:** a network of digitally-enabled last-mile access points (village-level entrepreneurs) delivering government and citizen services in rural and remote areas, the delivery backbone that makes many of the above (UMANG, DigiLocker, Aadhaar services) actually reachable for citizens without smartphones or internet literacy.

Ministry / Department	Technologies	Major Initiatives
Department of Biotechnology (DBT)	Genomics, microbiome science, biomanufacturing, vaccines, bio-foundries and Bio-AI Hubs	**BioE3 Policy (2024): Fostering high-performance biomanufacturing; Thematic areas:** bio-based chemicals, smart proteins, precision biotherapeutics, climate-resilient agriculture and marine/space biology through bio-AI hubs, bio-foundries and biomanufacturing hubs. **Genome India and Indian Human Microbiome Initiative:** 10,000-genome reference and population-level microbiome data hosted at the Indian Biological Data Centre; National One Health Mission for integrated human–animal–environment surveillance. **BIRAC** Acts as a single window for emerging biotech companies, startups, and researchers to bridge the gap between industry and academia **National Biopharma Mission:** Industry-Academia Collaborative Mission for accelerating biopharmaceutical development **INSACOG:** Indian SARS-CoV-2 Genomics Consortium (INSACOG), a multi-agency network coordinated by the Department of Biotechnology (DBT) to support Genomic surveillance
Department of Space (ISRO)	Launch vehicles, human spaceflight, Earth observation, navigation	**Gaganyaan:** First uncrewed flight and crewed flight targeted by 2027; Bharatiya Antariksh Station module BAS-01 approved, station operational by 2035; Chandrayaan-4 sample return. **Earth observation for governance:** NISAR and RISAT data; Bhuvan/NRSC crop, flood and disaster products; NavIC positioning; IN-SPACe enabling private launches and satellite data services for States.
Department of Atomic Energy (DAE)	Nuclear power, small modular reactors, radiation technologies	**Nuclear Energy Mission:** 100 GWe by 2047; at least five indigenous SMRs by 2033 (BSMR-200 and SMR-55 at Tarapur, high-temperature reactor for hydrogen at BARC Visakhapatnam); SHANTI Act 2025 opens the sector to private participation; installed capacity 8,780 MW rising to 22,380 MW by 2031–32. **Radiation technologies:** BARC food irradiation, radiopharmaceuticals and mutation-bred crop varieties.

Ministry / Department	Technologies	Major Initiatives
Ministry of Earth Sciences (MoES)	Weather radars and modelling, ocean technology, deep-sea vehicles	**Mission Mausam (₹2,000 crore):** expanded Doppler radar, wind-profiler and radiometer network, world's first floating radar (NIOT), Bharat Forecast System at 6 km resolution for district-level nowcasting and advisories. **Deep Ocean Mission / Samudrayaan:** Matsya-6000 crewed submersible trials, deep-sea mineral exploration and ocean-energy technologies.
Ministry of Agriculture & Farmers Welfare	Agricultural drones, remote sensing, genome-edited crops, soil sensing	**Namo Drone Didi (₹1,261 crore):** drone rental services by women SHGs for spraying; Kisan Drone SOPs and subsidies; Digital Crop Survey and satellite yield estimation under AgriStack. **ICAR crop science:** genome-edited rice varieties (DRR Dhan 100 Kamala, Pusa DST Rice 1), biofortified varieties, Clean Plant Programme; AgriMet district advisories with IMD.
Ministry of Rural Development / Dept of Land Resources	Drone survey, GIS, geo-tagging	**SVAMITVA:** drone survey of 3.29 lakh of 3.44 lakh villages and about 3.10 crore property cards (March 2026). **NAKSHA:** drone and aerial survey of urban habitations in pilot cities; DILRMP/ULPIN (Bhu-Aadhaar) linking cadastral maps, registration and revenue records; Bhuvan-based geo-tagging of MGNREGA assets.
Ministry of Housing & Urban Affairs	City sensors, SCADA, digital twins, geospatial master plans	**Smart Cities Mission:** Integrated Command and Control Centres in all 100 cities, backed by 83,000+ CCTV cameras; Intelligent Traffic Management Systems with Adaptive Traffic Control, Red Light Violation Detection and Automatic Number Plate Recognition; 17,000+ km of SCADA-monitored water networks; RFID-tracked waste fleets; city digital twins; GIS-based master plans under AMRUT 2.0.
Ministry of Health & Family Welfare / DHR / ICMR	Genomic surveillance, drone logistics, medical devices, diagnostics	**Surveillance:** INSACOG pathogen genomics; wastewater surveillance in cities linked to IDSP; i-DRONE delivery of vaccines and samples to remote areas. **Innovation:** ICMR/DHR AI-assisted screening (e.g., glaucoma, TB X-ray), indigenous medical devices under PLI, and mRNA and other vaccine platforms.

Ministry / Department	Technologies	Major Initiatives
Ministry of New & Renewable Energy	Green hydrogen, rooftop solar, storage, offshore wind	**National Green Hydrogen Mission (₹19,744 crore; 5 MMT by 2030):** 3,000 MW electrolyser manufacturing and 8.62 lakh tonnes per year of production capacity awarded; pilots in steel, mobility (37 vehicles, 9 stations) and shipping (green methanol bunkering at VOC Port). **PM Surya Ghar Muft Bijli Yojana:** 26 lakh rooftop installations by March 2026 against a target of one crore; PM-KUSUM solar pumps and feeder solarisation.
Ministry of Mines	Mineral exploration, processing and recycling technologies	**National Critical Mineral Mission (₹34,300 crore, seven years):** 230+ GSI exploration projects in 2025–26; four processing hubs; ₹1,500 crore recycling incentive (October 2025); up to 50 overseas assets by 2030.
Ministry of Civil Aviation	Drones and airspace management	**Drone Rules 2021 and Digital Sky:** amended 2022 to a self-certification, digital-authorisation model; ~90 per cent of airspace as Green Zone; 38,500+ registered drones; PLI for drone manufacturing; Drone Shakti and remote pilot training organisations. **DigiYatra:** a facial-recognition-based paperless entry system launched December 2022, using Aadhaar-linked, on-device biometric verification; live at 100+ airports with 75 million+ passenger journeys by December 2025.
Ministry of Road Transport & Highways / NHAI	Drone monitoring, new pavement materials, ITS	**Project monitoring:** mandatory monthly drone videography of highway works; use of steel slag, full-depth reclamation and geosynthetics; intelligent traffic and incident management on expressways. **FASTag:**RFID-based electronic toll collection, mandatory since 2021, now accounting for over 98 per cent of toll payments on National Highways; NHAI has proposed moving to fully cashless tolling (FASTag and UPI only) from April 2026.
Ministry of Railways	Signalling and traction technologies	**Kavach:** indigenous automatic train protection being rolled out across high-density routes; Vande Bharat and hydrogen-powered trains; Gati Shakti-based alignment planning.

Ministry / Department	Technologies	Major Initiatives
Ministry of Jal Shakti	Hydrological sensors, AI forecasting, water-quality monitoring	**CWC and Namami Gange:** AI/ML seven-day flood forecasting pilots; real-time water-quality monitoring stations on the Ganga; National Hydrology Project sensor and telemetry network; IoT-based monitoring of Jal Jeevan Mission schemes and Atal Bhujal groundwater.
Ministry of Environment, Forest & Climate Change	Satellite monitoring, air-quality sensing	**Forest and Air:** Forest Survey of India satellite forest-cover assessment and near-real-time fire alerts; CAAQMS sensor network under the National Clean Air Programme; Aravalli Green Wall.
Ministry of Heavy Industries / Petroleum & Natural Gas	Electric mobility, batteries, biofuels	**PM E-DRIVE and PLI-ACC (50 GWh):** electric buses, chargers and cell manufacturing; E20 ethanol blending achieved in 2025; SATAT compressed biogas and green hydrogen at refineries.
Ministry of Defence / DRDO	Deep-tech, autonomous systems, forecasting	**iDEX and ADITI:** funding for start-ups in drones, robotics and sensors; DRDO avalanche and weather forecasting using remote sensing and AI for Himalayan regions; Mission DefSpace.
Ministry of Home Affairs / NDMA	Surveillance, emergency response and early-warning systems	**Safe City Project:** launched 2018 under the Nirbhaya Fund, has sanctioned surveillance, panic-button and GIS crime-mapping infrastructure worth ₹2,919.55 crore across eight metro cities. **Emergency response:** the Emergency Response Support System, on the single number 112, integrates police, fire, health, disaster and women's safety response through a Public Safety Answering Point in each State and UT capital. **Disaster response:** National Database for Emergency Management (NDEM) geospatial platform; SACHET common alerting for cyclones, floods, heatwaves and lightning; NDRF drones and robotics for search and rescue. NDMA data show 58.6 per cent of India's landmass is vulnerable to moderate-to-severe earthquakes and 15 per cent to landslides and avalanches.